MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VIDICON

TYPE 7735A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic focus and deflection, 1 inch, for film or live pickup with color or black-and-white TV cameras

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	DIsj	isjT	Ec3	Ec2	Ec1	Ehk	FI	T (face-plate)	tk	I (focus coil)
Unit:	V	µAdc	µA	Vdc	Vdc	Vdc	Vdc	fc	°C	sec	mAdc
Maximum:	6.9	0.25	0.55	750	750	-300	-125	1,000	71	---	---
Minimum:	5.7	---	---	---	---	0	+10	---	0	---	---
TEST CONDITIONS:	6.3	Note 1	Note 2	Note 3	300	Note 4	---	Note 2	30 to 35	300	33

GENERAL:

Qualification - Required
Test pattern - EIA resolution chart, 1956, or equivalent (See note 5)
Scan size - See note 6
Test components - See note 7

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	25	Group D; t = 500 hours	---	---	---	---
---	Life-test end points:						
	Maximum grid No. 2 current	-		Ic2	900	---	µAdc

NOTES:

1. With zero incident light on tube face, the target voltage (Esj) shall be adjusted to obtain the specified dark current (DIsj) (Esj/DIsj), as specified under test conditions.
2. The faceplate illumination (FI) shall be adjusted to obtain specified target current (IsjT) as specified under test conditions. This may be done by adjustment of the lens opening. IsjT is defined as the sum of the signal-output current component (Isj) (0.20 µA) and the DIsj.
3. Grids No. 3 and No. 4 are connected together internally. The voltage applied to these two elements shall be referred to as Ec3. Ec3 voltages apply with the use of the specified focusing coil having a focus coil current of 33 mAdc, producing a field strength of approximately 40 Gauss. Ec3 and optical focus shall alternately be adjusted to obtain best focus (sharpest lines).
4. Adjust Ecl until the beam current is just sufficient to discharge the photo-layer as observed by presence of detail in the highlights of the monitor.
5. The specified test pattern shall be placed on a transparent slide and shall be illuminated from the rear of the slide.
6. Standard EIA (RETMA) television scan shall be employed. The scan size shall be 1/2 x 3/8 inch (12.70 x 9.53 mm).
7. Test components.
   (a) Deflection yoke - Cleveland Electronics No. VY-111-3, or equivalent.
   (b) Focusing coil - Cleveland Electronics No. VF-115-5, or equivalent.
   (c) Alinement coil - Cleveland Electronics No. VA-118, or equivalent.
   (d) The above components shall be mounted as shown on figure 3.
8. This test to be performed at the conclusion of the holding period.
9. (A) The tube shall be tested in camera with associated monitors as shown in method 5426.
10. Target quick age.
    (a) With no light on the tube, adjust horizontal, and vertical camera deflection, Ec3 and focus coil current to just overscan the target.
    (b) Set the target voltage at 300 volts and slightly defocus by adjusting Ec3. After 300 volts a lower target voltage is permissible if burn can be removed.
    (c) Adjust Ecl to obtain a complete white circular area on monitor with no distortion or dark areas present. The target current should read between 1 and 3 µAdc.
    (d) Age traget for a minimum of 1 minute if target burns are encountered.

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17. -Continued

(d) Maximum concentration of spots shall be evaluated by one of the three methods shown below:

Method I:	Minimum separation between any 2 spots greater than 0.015 inch (0.38 mm) limited to 1/4 inch (6.35 mm).
Method II:	Minimum separation between any 2 spots greater than 0.19 percent of raster height is limited to a distance equivalent to 3.1 percent of raster height.
Method III:	Minimum separation between any 2 spots greater than 1 raster line is limited to 16 lines.

(e) Tubes are rejected for smudge, lines, streaks, mottled background, grainy background, or uneven background having contrast ratios greater than 1.5 to 1.0.

18. Recheck Esj after alinement is completed.

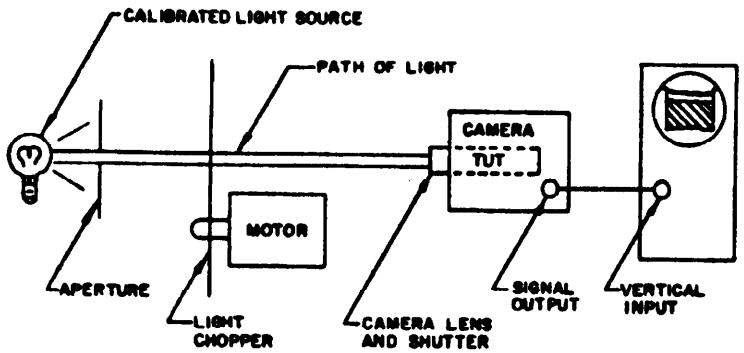
19. Project specified faceplate illumination on vidicon target. The lamp current of the light source shall be adjusted for a color temperature of 2,856°K. Sensitivity is the target current under these conditions minus dark current.

20. Lag test.

(a) Equipment:

(1) Calibrated light source of 2,856°K color temperature.
(2) Suitable lens and apertures to obtain specified pattern and faceplate illumination.
(3) Motor-driven light chopper capable of interrupting the path of light 4 times per second. The light chopper shall provide light to the vidicon target for 1/5 second and interrupt for 1/20 second such that the vidicon target receives light for about 12 TV fields and receives no light for about 3 TV fields.
(4) An oscilloscope capable of monitoring the signal output including the dc component. (P7 screen is recommended.)

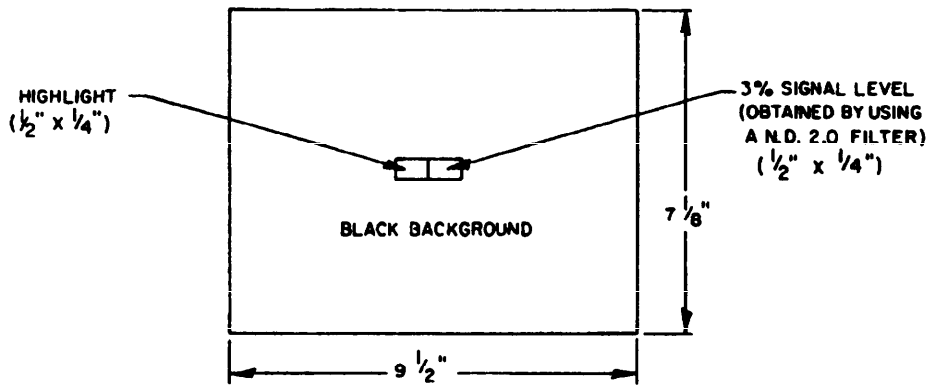
(b) Block diagram:



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23. Using the spurious signal test pattern as shown in note 11, set up tube to specified conditions. Replace above test pattern by the "After image test pattern" as shown below. The highlight brightness of the after image test pattern shall be the same as the highlight brightness of the spurious signal test pattern. Line select the waveform monitor trace through the center information. (Use IEEE response to reduce noise.) Adjust gain to make peak black-to-white signal level equal to 100 percent. Using the step gain switch, increase the signal amplitude to where a 3 percent signal level can be clearly seen. Expose the vidicon to the test pattern for an exact 30 seconds. Cap up the tube. Record the time (in seconds, using a stopwatch) required for the residual peak white signal to decay to a 3 percent level (above black) as after image.



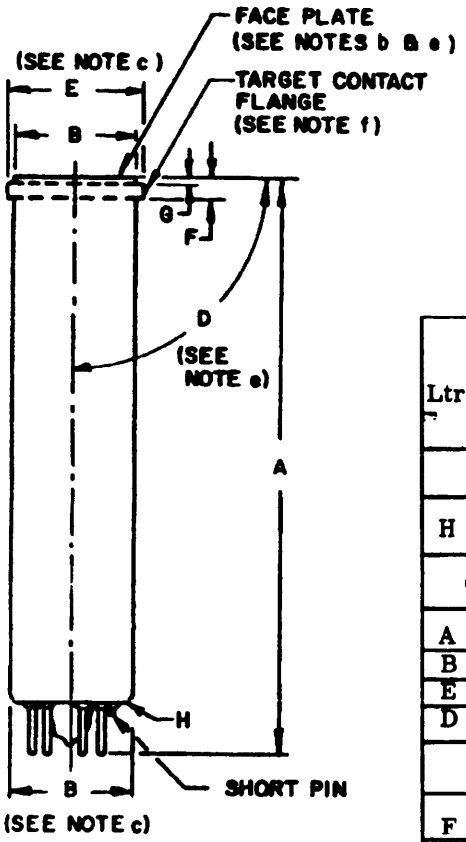
24. Raster burn test.

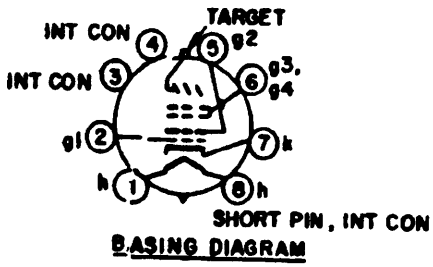
(a) With conditions as shown for 1/2 x 3/8-inch (12.70 x 9.53 mm) scan, completely overscan target and increase bias until complete target is discharged.
(b) Set black level at a value where no part of the dark current signal is clipped.
(c) Increase monitor bias until complete target is just visible.
(d) If a raster burn or image burn is present, repeat target quick age test.
(e) Repeat steps (a), (b), and (c).
(f) Reject if raster burn is lighter than surrounding area. Reject for any permanent image burn.

25. Life-test conditions.

(a) Adjustment for scan size is made by the use of a specially designed vidicon tube with a phosphor screen and with 300 Vdc applied to the target. Use RCA deflection yoke No. 208007, or equivalent, focusing coil No. 208005, or equivalent, and alinement coil No. 208006, or equivalent.
(b) With the scan size adjusted to approximately 1/2 x 3/8 inch (12.70 x 9.53 mm) (do not allow scanned area to touch target ring), faceplate temperature of 40°±5°C, adjust Ec1 to obtain 0.5 mAdc of grid No. 2 current. Cover left half of faceplate (looking into tube face with key pin up) with black tape, and adjust target voltage to obtain 0.20 uAdc (DIsj) with no light incident on the faceplate.
(c) Adjust the incident illumination on the faceplate to obtain 0.40 uAdc target current (IsjT). (Light plus dark current.)
(d) The target voltage shall be readjusted for specified dark current within 24 hours after the tube has reached the specified operating temperature. The incident illumination shall be readjusted when Esj is readjusted.

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BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
H	Base: E8-11 (see note d)	
Quality conformance inspection, part 2		
A	6.125 (155.58)	6.375 (161.93)
B	0.990 (25.15) Dia.	1.045 (26.54) Dia.
E	1.116 (28.35) Dia.	1.132 (28.75) Dia.
D	89.75°	90.00°
Reference dimensions (see note a)		
F	0.175 (4.45)	
G	0.050 (1.27)	

NOTES:
a. These dimensions are for information only and are not required for inspection purposes.
b. Direction of incident light is into face end of tube.
c. Concentricity and camber of tube shall be determined by gage shown on figure 2.
d. JEDEC designation. For pin alinement, use JEDEC gage GE8-2.
e. Faceplate tilt shall be gaged about entire circumference.
f. Target contact flange in form of metal ring, and with diameter as specified for dimension E, may be located along tube axis in any part of, or all of, space between dashed lines.

FIGURE 1. Outline drawing of electron tube type 7735A.

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